

Assignment 1: Parallel Order Book Processing with OpenMP

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Part 1: Snapshot Generation

In this part we decouple *packet processing* from *snapshot processing*.

A single thread processes packets sequentially and updates the live order book state, maintaining the most recent buy and sell values per stock. At snapshot boundaries, the thread copies the active stock data and dispatches as an OpenMP task. Snapshot tasks operate sort the stocks by spread, and write output files independently.

Part 2: Order Book Statistics

Total Amount Traded

The total traded amount is computed as the sum of $\text{orderQty} \times \text{orderValue}$ over all packets. This computation is implemented using an OpenMP `reduction` clause.

Per-Stock Statistics

To compute per-stock statistics we use thread-local aggregation. Each thread maintains private statistics for each stock ID while scanning a subset of packets. After packet processing, a parallel reduction phase merges thread-local data into global results, which is then written to disk.

Performance Evaluation

Performance was evaluated by varying the number of threads which resulted in the following graph.

